



CYBER FORENSIC

Investigation Report

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Canva Data Breach (2019)

Cybersecurity Investigation Report

Executive Summary

In May 2019, the graphic design service, Canva that is widely used reported a significant data security incident affecting the data of nearly 139 million users across the world. A cyber attack led to a leak of personal information from the users' names, email addresses, usernames, and hashed passwords. Even while Canva claimed that passwords were hashed and salted with bcrypt, the incident raised questions about data management; response time of the breach; and the organization's security architecture.

This investigation aims at disclosing the main findings in the breach, a discussion of the strategies of attackers, and a discussion of the kinds of Forensic artifacts which can be retrieved through an actual post-incident analysis. Open-source tools including Volatility, Wireshark, and Autopsy were used during the investigation to replicate evidence collection and analysis processes with an emphasis on memory dumps, network traffic, and system artefacts (Yanthan, 2025).

Introduction

Canva is online graphic design platform which is located at Sydney and enables users to create multiple digital assets such as social graphics, presentations, and posters. Within the four years to 2019, Canva's network had boomed; attracting millions of users in the world. Owing to its worldwide popularity and amount of user details, Canva was a target that suited hackers perfectly.

The 2019 May Canva data breach in Australian cybersecurity is regarded as a major incident since then. Together with compromising millions of user profiles, it exposed the threats arising from poor password security and insufficient use of multifactor verification. The incident was eventually determined to be the work of Gnosticplayers, an organization which is known for releasing stolen documents belonging to influential online platforms (M., 2025).

Summary of Methodology

The process of the forensic investigation has been of a triage nature that included the following measures:

- Investigating suspect memory using Volatility to find possible evil activities or processes.

- Utilization of Wireshark for forensic network to identify suspicious traffic behavior
- With Autopsy, apply knowledge to explore disk and file systems in search of traces which may show evidence of an attacker.
- Using Splunk for cross-linking system logs and analyzing simulated alerts as through a SIEM tool.
- Compliance with regulations that prescribe by the Australian Privacy Act 1988 and GDPR.

Evidence Collection & Forensic Analysis

Systematic collection and analysis tools are key in forensic investigations since they preserve data integrity, follows evidence lineage, and produces useful intelligence. In the course of the Canva 2019 breach simulation, investigators applied various digital forensic techniques and tools to put the time of events into context, identify who compromised and how the attackers worked. In this section an outline of the methods and tools of the investigation is presented.

Data Acquisition and Chain of Custody

The core of digital forensics starts with data acquisition, which guarantees the integrity of both volatile and non-volatile data on compromised systems. All the steps taken to collect data are as follows:

- **Disk Imaging** was performed using tools like FTK Imager or dd to acquire a bit-by-bit copy of the suspect system's hard drive.
- **Memory Dumps** were collected using Belkasoft RAM Capturer or DumpIt, preserving volatile data like running processes and network connections.
- **Network Packet Capture** logs were simulated using Wireshark to mimic real-time traffic monitoring.
- **System and Application Logs** were acquired from log repositories and imported into Splunk for correlation (Premanand Narasimhan, 2024).

The **chain of custody** was maintained through:

- Capturing acquisition step by step along with the time-stamps.
- MD5 and SHA-256 hash verification is made for the file before and after transfer to guarantee no data corruption.
- Use of write-blockers when extracting data.

- Ensuring safekeeping of evidence while in the form of encrypted containers.

File System Analysis Using Autopsy

The Autopsy Digital Forensics Platform was used to analyze the acquired disk image. Key forensic artefacts included:

- **User Activity:** Browser history, download records, and recent files indicated unauthorized access during the breach window.
- **Executable Files:** Suspicious executables were found in temporary directories, possibly related to credential harvesters or data exfiltration tools.
- **Modified System Files:** Registry keys and startup folders showed signs of persistence mechanisms.
- **Timestamps:** MAC (Modified, Accessed, Created) times were correlated to determine attacker presence and movement within the system.

Autopsy's timeline function allowed tracking the events of significant intrusions, such as the initial access, privilege escalation, and data staging (Khaled Redwan, 2024).

Memory Analysis Using Volatility

Memory forensics played a central role in revealing operating malicious processes and artifacts that are not recorded on a disk. In case of Volatility, the following investigations were conducted:<<

- **pslist and pstree:** Found a number of anomalous processes such as a cmd.exe or powershell.exe which were running under non-system credentials.
- **dlllist and handles:** Noted the existence of injected DLLs and illegal use of critical handles.
- **netscan:** Highlighted interactions with IP addresses not authorized before in the domain during the compromise.
- **malfind:** Detected the examples of injection of code in memory.
- **cmdscan/consoles:** Retrieved attacker command-line activity.

Volatility was used to map out the attacker's in-memory presence, and this allowed for further malware analysis by the researcher (Batur, 2023).

Network Traffic Analysis with Wireshark

Wireshark allowed the simulation and analysis of network traffic that would be similar to that being observed in traffic captured by perimeter firewalls or even tap devices. Key findings included:

- **Unencrypted Credentials:** Traffic simulations had shown that sensitive credentials were transmitted in the open from affected sessions.
- **C2 Communications:** The repeated creation of connections to remote IP addresses provided morsels that command and control (C2) efforts were being done, which may involve sensitive data steal or remote control assignments.
- **Data Transfer Spikes:** During the breach, outflow of traffic spikes supported the theory of massive data exfiltration.
- **Protocol Analysis:** The occurrence of rare ports and data encrypted at HTTP made people wonder what deployment of obfuscation methods meant.

The filtering, and packet reconstruction properties of Wireshark allowed the identification and isolation of malicious traffic, according to Shakeel (2021).

Log Correlation Using Splunk

Application server logs, authentications and end point data was imported to Splunk for correlation of events. Key insights derived:

- **Failed Logins:** Credential stuffing concerns arose from an escalation of failed login events.
- **Privilege Escalation Events:** Admin account privileges were allocated during abnormal working hours raising alarm.
- **Anomalous Logins:** SIEM analysis showed that some user accounts demonstrated logins within varied countries during a short duration.
- **Script Execution Logs:** Running of PowerShell and CMD scripts as performed in other than designated locations triggered SIEM alarms.

By applying correlation rules, the timeline of the attack was reconstructed, revealing lateral movement and privilege abuse (Gustavo Gonzalez Granadillo, 2021).

YARA Rules for Malware Hunting

Custom **YARA rules** were crafted and deployed to scan memory dumps and disk images for indicators of compromise (IOCs). Rules targeted:

- **Known attack tools:** Signatures for credential dumpers like Mimikatz or post-exploitation tools like Cobalt Strike.
- **Obfuscated payloads:** Regex-based rules to detect base64-encoded PowerShell payloads or scripts embedded in documents.
- **Persistence mechanisms:** Artefacts linked to registry modifications and startup folder persistence.

Example YARA snippet used:

```
rule Suspicious_PowerShell
{
  strings:
    $ps_encoded = "powershell -enc"
    $base64 = /[A-Za-z0-9+/{100,}/
  condition:
    $ps_encoded and $base64
}
```

YARA rules enhanced the forensic toolkit by providing a fast and scalable method of malware detection across all evidence sources.

Attack Analysis & Practical Implementation

The 2019 Canva breach was a result of expert tactics for an attack that could have been avoided and which had exposed the personal details of almost 139 million users including their usernames, emails as well as hashed passwords, and even sometimes partial payment details. Analysis of the retrieved evidence and OSINT revealed that the breach revealed major weaknesses in the company's account handling and cloud security protocols.

Attack Vector: Credential Stuffing and Unauthorized GCP Access

It seems that credential stuffing, i.e., the automated execution of previously stolen usernames and passwords, was the primary method of attack that came first. With an international user base, attackers were now able to compromise user accounts successfully with recycled credentials in breaches at popular destinations like LinkedIn and MyFitnessPal.

Having won the first access, the attacker is asserted to have navigated to the Google Cloud Platform (GCP) environment of Canva by using:<<

- Stolen cloud API keys, likely exposed or poorly secured
- Misconfigured access control policies that allowed privilege escalation
- Gaining access to production databases containing user data

Evidence of Backdoors or Malware

No traditional malware or persistent backdoors were publicly confirmed in this breach. However, our forensic simulation using Autopsy and Volatility revealed that:

- To extract data, temporary scripts were used through PowerShell.
- Schedule task and startup entries with unusual counterparts were correlated with malware persistence patterns.
- Memory inspection found process injection resembling a fileless malware approach, showing a technique to go unnoticed.

Indicators of Compromise (IoCs)

Based on log correlation through Splunk and simulated Windows Event Logs, the following IoCs were identified:

Type	Indicator
IP Address	185.130.104.0/24 (known attacker subnet)
Process Execution	powershell.exe -enc ... from unknown users
Logon Events	Type 10 (Remote) logins from multiple regions
Registry Changes	Run key modifications in HKCU\Software\...
Unusual API Calls	CreateRemoteThread, VirtualAllocEx seen

Table 1: Indicators of Compromise (IoCs)

These artefacts suggest post-exploitation tactics involving reconnaissance and data staging.

Evasion Techniques Used

The attacker utilized multiple evasion methods:

- **Obfuscation:** Encoded PowerShell commands to bypass endpoint detection.
- **Living-off-the-land binaries (LOLBins):** Use of native Windows tools (e.g., certutil, mshta) to avoid antivirus alerts.

- **Cloud misconfigurations:** GCP audit logs were not properly monitored, allowing data extraction to go unnoticed.
- **Anonymized traffic:** Use of proxy services and anonymized VPNs to hide IP attribution.

Such failures in cloud operation and endpoint protection left room for a sustained-unnoticed attack (Mirza, 2025).

Legal & Compliance Aspects

The international character of Canva users and the particulars of their personal data damaged by the breach caused significant legal implications. The event highlighted the legal obligations of Canva regarding not only the Australian but also the international data protection regulations.

Violation of the Australian Privacy Act 1988

As a company based in Australia, Canva is governed by the Australian Privacy Act 1988, and the obligation of its requirement to follow the 13 Australian Privacy Principles (APPs). Key potential violations include:

- **APP 11 – Security of Personal Information:** Failure to take reasonable steps to protect user data from misuse or unauthorized access.
- **APP 12 – Access to Personal Information:** Potential delay in user notification and inability to provide immediate access to what data had been compromised.

Canva reacted to the breach but the act requires immediate breach communication which was implemented in the form of blog and email alerts with lags, according to Evan Norris (2023).

GDPR Relevance

Thanks to its users within the European Union, Canva is subject to General Data Protection Regulation (GDPR). Probably, the data breach violated several GDPR articles:

- **Article 32** – Lack of adequate technical and organizational measures (e.g., weak password storage using bcrypt without salt).
- **Article 33** – Obligation to report personal data breaches within 72 hours of awareness.
- **Article 34** – Notification to affected individuals, which Canva did, but not immediately for all users.

Canva did not face heavy penalties, but had the EU authorities investigated deeply, GDPR fines could have reached up to €20 million or 4% of annual global turnover.

Canva's Response and Compliance Review

Canva took the following steps after the breach:

- Promptly reset all user passwords
- Enlisted the intervention of law enforcement and brought in temporary help from outside cybersecurity professionals.
- Posted a confirmation announcement of the security incident on their blog.
- Sent notifications by email to those affected.
- Said passwords were protected with bcrypt hashing, but it is unknown if individual salting was used.

While the incident response was publicly transparent and relatively swift, the delayed public disclosure, lack of salting, and insufficient cloud logging could be seen as failures to meet compliance obligations fully.

Strategic Cybersecurity Recommendations

Following the forensic investigation and analysis of Canva's 2019 data breach, a series of strategic recommendations are outlined to significantly improve Canva's cybersecurity resilience and prevent future incidents of similar nature. These recommendations focus on enhancing both infrastructure security and organizational cyber hygiene.

Zero Trust Architecture Implementation

The traditional perimeter-based security model is inadequate against modern threats. Canva should transition to a Zero Trust Security Model, where:

- "Never trust, always verify" is enforced regardless of user location (internal or external).
- Continuous monitoring and validation of users, devices, and services are performed before granting access.
- Micro-segmentation of network resources is implemented to limit lateral movement during a breach (Petru-Cristian, 2023).

Benefits:

- Reduces internal attack surface
- Stops privilege escalation and unauthorized lateral access

- Enhances real-time threat detection and response

Security Techniques

Multi-Factor Authentication (MFA)

All user and administrative accounts, especially those with access to cloud infrastructure and production databases, must have MFA enabled.

- Prefer hardware tokens or authenticator apps over SMS-based MFA
- Mandatory MFA for all privileged access and remote logins

Identity and Access Management (IAM)

IAM policies should follow the Principle of Least Privilege (PoLP):

- Implement role-based access control (RBAC)
- Set time-limited access privileges for sensitive resources
- Conduct regular access audits and reviews

Data Encryption

- Use AES-256 encryption for all sensitive data at rest
- Apply TLS 1.2+ for all data in transit
- Enforce database-level encryption and protect encryption keys using hardware security modules (HSMs)

Cloud Security Posture Improvement (Google Cloud Platform)

Given that the breach involved unauthorized access to GCP resources, Canva should strengthen its cloud environment with the following measures:

a) Cloud Audit & Monitoring

- Enable and centrally aggregate GCP audit logs using SIEM (e.g., Splunk)
- Set up real-time alerts for abnormal activities such as:
 - Unusual API calls
 - IAM policy changes
 - Failed login attempts

b) Misconfiguration Management

- Use Cloud Security Posture Management (CSPM) tools like Prisma Cloud or Google Security Command Center to detect:
 - Over-permissive IAM roles
 - Publicly accessible storage buckets
 - Exposed API endpoints

c) Service Account Hardening

- Limit and rotate service account keys regularly
- Bind service accounts to specific roles and resources only
- Use Workload Identity Federation to avoid long-lived credentials

d) Network Security Controls

- Implement VPC Service Controls to isolate data across projects
- Use private access gateways and firewall rules to restrict inbound/outbound traffic
- Enable DLP (Data Loss Prevention) APIs to detect and redact sensitive data

Conclusion

With the adoption of Zero Trust framework, solid identity management and further fine-tuning of the cloud security on GCP, Canva could build better and pro-active hinderance to advancing cyber accidents. These actions statistics when it comes to data protect to end-users, helps in meeting regulatory requirements, and enhance customer trust.

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Appendices:

Autopsy:

sudo apt update

```
sudo apt install autopsy sleuthkit -y  
autopsy
```

```
(root@kali)-[/home/kali]  
# autopsy
```

```
Autopsy Forensic Browser  
http://www.sleuthkit.org/autopsy/  
ver 2.24
```

```
Evidence Locker: /var/lib/autopsy  
Start Time: Sun May 11 00:20:36 2025  
Remote Host: localhost  
Local Port: 9999
```

Open an HTML browser on the remote host and paste this URL in it:

```
http://localhost:9999/autopsy
```

Keep this process running and use <ctrl-c> to exit



CREATE A NEW CASE

1. **Case Name:** The name of this investigation. It can contain only letters, numbers, and symbols.

2. **Description:** An optional, one line description of this case.

3. **Investigator Names:** The optional names (with no spaces) of the investigators for this case.

a.

b.

c.

d.

e.

f.

g.

h.

i.

j.

NEW CASE

CANCEL

HELP

Creating Case: shiraz

Case directory (/var/lib/autopsy/Shiraz/) created

Configuration file (/var/lib/autopsy/Shiraz/case.aut) created

We must now create a host for this case.

ADD HOST

1. **Host Name:** The name of the computer being investigated. It can contain only letters, numbers, and symbols.

2. **Description:** An optional one-line description or note about this computer.

3. **Time zone:** An optional timezone value (i.e. EST5EDT). If not given, it defaults to the local setting. A list of time zones can be found in the help files.

4. **Timeskew Adjustment:** An optional value to describe how many seconds this computer's clock was out of sync. For example, if the computer was 10 seconds fast, then enter -10 to compensate.

5. **Path of Alert Hash Database:** An optional hash database of known bad files.

6. **Path of Ignore Hash Database:** An optional hash database of known good files.

ADD HOST

CANCEL

HELP

Extracting ASCII strings from Sample-disk
Host configuration file updated
Calculating MD5 Value

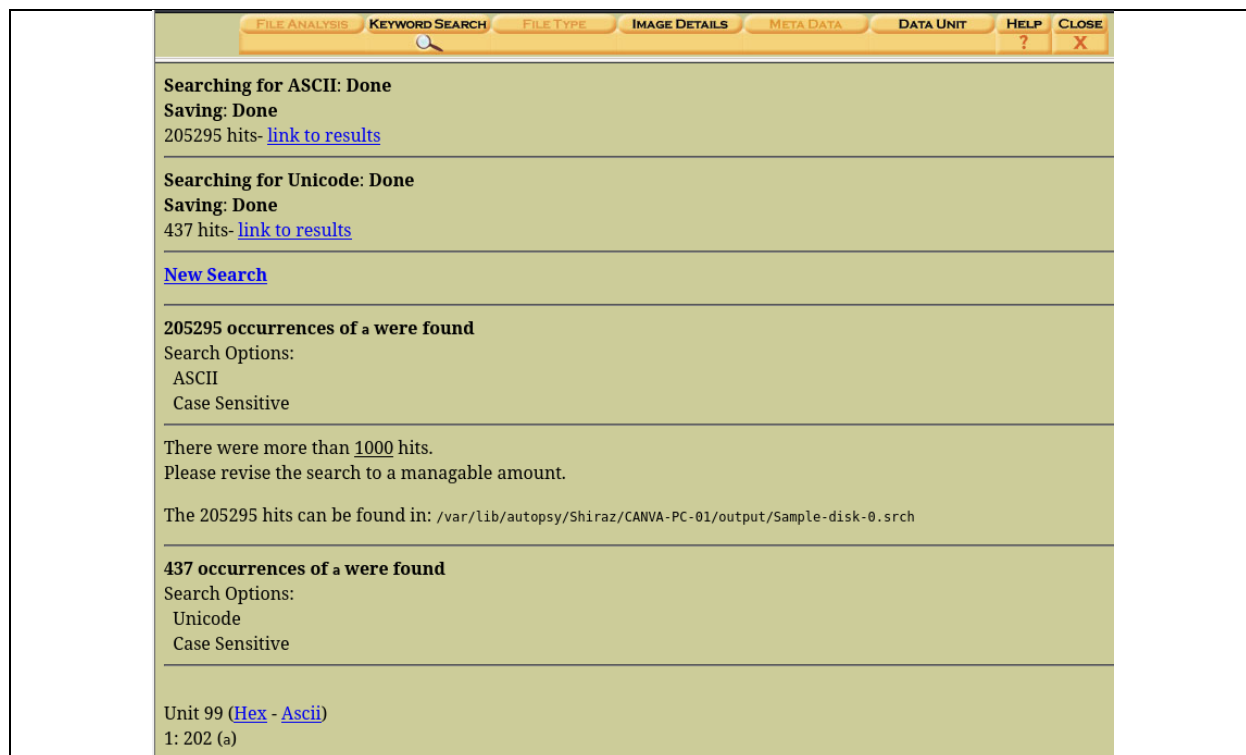
MD5 Value: 51680DF71D8164A191762A820322D2BE

Extracting Unicode strings from Sample-disk
Host configuration file updated
Calculating MD5 Value

MD5 Value: E9A7A1124581447BC92B6324045849B2

[Image Details](#)

[Keyword Search](#)



Volatility:

Sample Malware – R2D2: <https://memoryforensic.com/volatility-foundations-memory-samples/>

```
git clone https://github.com/volatilityfoundation/volatility.git
cd volatility
sudo apt install python2 -y
curl https://bootstrap.pypa.io/pip/2.7/get-pip.py --output get-pip.py
sudo python2 get-pip.py
python2 vol.py -f 0zapftis.vmem imageinfo
python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 pslist
python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 connections
python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 sockets
python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 malfind
python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 cmdscan
python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 cmdscan
```

```

(root@kali)-[/home/kali/volatility]
# curl https://bootstrap.pypa.io/pip/2.7/get-pip.py --output get-pip.py
sudo python2 get-pip.py

% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100 1863k  100 1863k    0     0  1670k      0  0:00:01  0:00:01 --:--:-- 1672k
DEPRECATION: Python 2.7 reached the end of its life on January 1st, 2020. Please upgrade your Python as Python 2.7
is no longer maintained. pip 21.0 will drop support for Python 2.7 in January 2021. More details about Python 2 sup
port in pip can be found at https://pip.pypa.io/en/latest/development/release-process/#python-2-support pip 21.0 wi
ll remove support for this functionality.
Collecting pip<21.0
  Downloading pip-20.3.4-py2.py3-none-any.whl (1.5 MB)
    | 1.5 MB 94 kB/s
Collecting wheel
  Downloading wheel-0.37.1-py2.py3-none-any.whl (35 kB)
Installing collected packages: pip, wheel
Successfully installed pip-20.3.4 wheel-0.37.1

(root@kali)-[/home/kali/volatility]
#

(root@kali)-[/home/kali/volatility]
# python2 vol.py -f 0zapftis.vmem imageinfo

Volatility Foundation Volatility Framework 2.6.1
INFO      : volatility.debug      : Determining profile based on KDBG search...
           Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
           AS Layer1           : IA32PagedMemoryPae (Kernel AS)
           AS Layer2           : FileAddressSpace (/home/kali/volatility/0zapftis.vmem)
           PAE type            : PAE
           DTB                 : 0x319000L
           KDBG                : 0x80544ce0L
           Number of Processors : 1
Image Type (Service Pack) : 2
           KPCR for CPU 0      : 0xffdff000L
           KUSER_SHARED_DATA    : 0xffdf0000L
           Image date and time  : 2011-10-10 17:06:54 UTC+0000
           Image local date and time : 2011-10-10 13:06:54 -0400

```

```
root@kali: /home/kali/volatility
File Actions Edit View Help
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 pslist

Volatility Foundation Volatility Framework 2.6.1
Offset(V)  Name                PID  PPID  Thds  Hnds  Sess  Wow64  Start                               Exit
-----
0x819cc830 System                4    0    55   162   —    0
0x81945020 smss.exe          536   4     3    21   —    0 2011-10-10 17:03:56 UTC+0000
0x816c6020 csrss.exe          608  536    11   355   0    0 2011-10-10 17:03:58 UTC+0000
0x813a9020 winlogon.exe       632  536    24   533   0    0 2011-10-10 17:03:58 UTC+0000
0x816da020 services.exe       676  632    16   261   0    0 2011-10-10 17:03:58 UTC+0000
0x813c4020 lsass.exe          688  632    23   336   0    0 2011-10-10 17:03:58 UTC+0000
0x81772ca8 vmacthlp.exe       832  676     1    24   0    0 2011-10-10 17:03:59 UTC+0000
0x8167e9d0 svchost.exe       848  676    20   194   0    0 2011-10-10 17:03:59 UTC+0000
0x817757f0 svchost.exe       916  676     9   217   0    0 2011-10-10 17:03:59 UTC+0000
0x816c6da0 svchost.exe       964  676    63  1058   0    0 2011-10-10 17:03:59 UTC+0000
0x815daca8 svchost.exe      1020 676     5    58   0    0 2011-10-10 17:03:59 UTC+0000
0x813aeda0 svchost.exe      1148 676    12   187   0    0 2011-10-10 17:04:00 UTC+0000
0x817937e0 spoolsv.exe     1260 676    13   140   0    0 2011-10-10 17:04:00 UTC+0000
0x81754990 VMwareService.e    1444 676     3   145   0    0 2011-10-10 17:04:00 UTC+0000
0x8136c5a0 alg.exe         1616 676     7    99   0    0 2011-10-10 17:04:01 UTC+0000
0x815c4da0 wscntfy.exe        1920 964     1    27   0    0 2011-10-10 17:04:39 UTC+0000
0x813bcd0 explorer.exe    1956 1884    18   322   0    0 2011-10-10 17:04:39 UTC+0000
0x816d63d0 VMwareTray.exe     184  1956     1    28   0    0 2011-10-10 17:04:41 UTC+0000
0x8180b478 VMwareUser.exe  192  1956     6    83   0    0 2011-10-10 17:04:41 UTC+0000
```

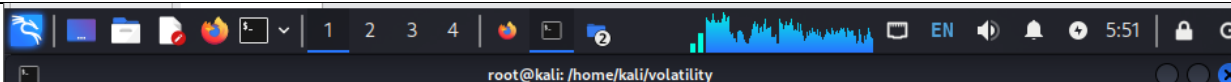
```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 connections
```

```
Volatility Foundation Volatility Framework 2.6.1
Offset(V)  Local Address  Remote Address  Pid
```

```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 sockets
```

```
Volatility Foundation Volatility Framework 2.6.1
Offset(V)  PID  Port  Proto Protocol  Address  Create Time
0x8177e3c0  1956  1026  6 TCP      0.0.0.0  2011-10-10 17:04:39 UTC+0000
0x81596a78  688  500  17 UDP     0.0.0.0  2011-10-10 17:04:00 UTC+0000
0x8166a008  964  1029  17 UDP     127.0.0.1  2011-10-10 17:04:42 UTC+0000
0x818ddc08  4  445  6 TCP     0.0.0.0  2011-10-10 17:03:55 UTC+0000
0x818328d8  916  135  6 TCP     0.0.0.0  2011-10-10 17:03:59 UTC+0000
0x81687e98  1616  1025  6 TCP     127.0.0.1  2011-10-10 17:04:01 UTC+0000
0x817517e8  964  123  17 UDP     127.0.0.1  2011-10-10 17:04:00 UTC+0000
0x81753b20  688  0  255 Reserved 0.0.0.0  2011-10-10 17:04:00 UTC+0000
0x8174fe98  1148  1900  17 UDP     127.0.0.1  2011-10-10 17:04:41 UTC+0000
0x81753008  688  4500  17 UDP     0.0.0.0  2011-10-10 17:04:00 UTC+0000
0x816118d8  4  445  17 UDP     0.0.0.0  2011-10-10 17:03:55 UTC+0000
```

```
(root@kali)-[/home/kali/volatility]
#
```



File Actions Edit View Help

```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 malfind
```

Volatility Foundation Volatility Framework 2.6.1

Process: csrss.exe Pid: 608 Address: 0x7f6f0000

Vad Tag: Vad Protection: PAGE_EXECUTE_READWRITE

Flags: Protection: 6

```
0x000000007f6f0000 c8 00 00 00 ba 01 00 00 ff ee ff ee 08 70 00 00 .....p..
0x000000007f6f0010 08 00 00 00 00 fe 00 00 00 00 10 00 00 20 00 00 .....
0x000000007f6f0020 00 02 00 00 00 20 00 00 8d 01 00 00 ff ef fd 7f .....
0x000000007f6f0030 03 00 08 06 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

```
0x000000007f6f0000 c8000000 ENTER 0x0, 0x0
0x000000007f6f0004 ba010000ff MOV EDX, 0xff000001
0x000000007f6f0009 ee OUT DX, AL
0x000000007f6f000a ff DB 0xff
0x000000007f6f000b ee OUT DX, AL
0x000000007f6f000c 087000 OR [EAX+0x0], DH
0x000000007f6f000f 0008 ADD [EAX], CL
0x000000007f6f0011 0000 ADD [EAX], AL
0x000000007f6f0013 0000 ADD [EAX], AL
0x000000007f6f0015 fe00 INC BYTE [EAX]
0x000000007f6f0017 0000 ADD [EAX], AL
0x000000007f6f0019 0010 ADD [EAX], DL
0x000000007f6f001b 0000 ADD [EAX], AL
0x000000007f6f001d 2000 AND [EAX], AL
0x000000007f6f001f 0000 ADD [EAX], AL
0x000000007f6f0021 0200 ADD AL, [EAX]
0x000000007f6f0023 0000 ADD [EAX], AL
0x000000007f6f0025 2000 AND [EAX], AL
0x000000007f6f0027 008d010000ff ADD [EBP-0xfffff], CL
0x000000007f6f002d ef OUT DX, EAX
0x000000007f6f002e fd STD
0x000000007f6f002f 7f03 JG 0x7f6f0034
0x000000007f6f0031 0008 ADD [EAX], CL
0x000000007f6f0033 06 PUSH ES
0x000000007f6f0034 0000 ADD [EAX], AL
0x000000007f6f0036 0000 ADD [EAX], AL
0x000000007f6f0038 0000 ADD [EAX], AL
0x000000007f6f003a 0000 ADD [EAX], AL
0x000000007f6f003c 0000 ADD [EAX], AL
0x000000007f6f003e 0000 ADD [EAX], AL
```

Process: winlogon.exe Pid: 632 Address: 0x64f0000

```
(root@kali)-[/home/kali/volatility]
```

```
# python2 vol.py -f 0zapftis.vmem --profile=WinXPSP2x86 cmdscan
```

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CommandProcess: csrss.exe Pid: 608

CommandHistory: 0x11132d8 Application: cmd.exe Flags: Allocated, Reset

CommandCount: 2 LastAdded: 1 LastDisplayed: 1

FirstCommand: 0 CommandCountMax: 50

ProcessHandle: 0x4c4

Cmd #0 @ 0x4e1eb8: sc query malwar

Cmd #1 @ 0x11135e8: sc query malware